

Lagrangian relaxation for accumulated metric constraints

February 20, 2026

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1 Ambient condition statistics

We assume that the statistics of ambient conditions are described in the form of a probability mass function (PMF)

$$p_A(a_n) = P(A = a_n), \quad n = 1, \dots, N \quad (1)$$

where A is a discrete random variable denoting the ambient condition. We will use

$$\mathbf{a} := (a_1, \dots, a_N), \quad (2)$$

$$\begin{aligned} \mathbf{p} &:= (p_1, \dots, p_N) \\ &:= (p_A(a_1), \dots, p_A(a_N)) \end{aligned} \quad (3)$$

to denote the set of all considered discrete ambient conditions and their corresponding probabilities.

Remarks

- Of course, the discrete PMF is an approximation of an underlying continuous probability density function over random ambient conditions. In practice, any observed ambient condition will be assigned to one of the discrete ambient conditions (the “closest” one, ideally).
- All probability distributions are assumed stationary and ergodic, so that $\mathbf{p}$ does not change over time.
- We use lower-case a to denote a single ambient condition, although the following considerations equally hold for multi-dimensional ambient conditions (e.g. wind speed, wind direction, electricity price, ...).

2 Plant model

We assume steady-state plant models f of the form

$$y = f(a, u), \quad (4)$$

where a denotes an ambient condition according to Section 1 and u denotes a control setpoint.

Remarks

- Unlike the discrete set of ambient conditions, the control input u and the plant model f are assumed continuous.
- Only steady-state, time-invariant models are considered, so no internal dynamics or time-dependent behavior are modelled.
- Similar to a we use lower-case u to denote a control input, although the control can be multi-dimensional (yaw-steering, power regulation, etc. for multiple turbines, ...). We will refer to a specific set of control inputs as a control setpoint (although this may in fact refer to a set of setpoints).

3 Control policy

We are interested in control policies

$$u = \pi(a|\boldsymbol{\theta}) \quad (5)$$

that assign to any ambient condition a a control input u , and that are parametrizable by a set of parameters $\boldsymbol{\theta}$. If we limit a to the discrete set of ambient conditions $\mathbf{a}$ from Section 1, a *discrete* control policy is specified by a set of discrete control setpoints

$$\mathbf{u}^\pi(\boldsymbol{\theta}) = (u_1^\pi(\boldsymbol{\theta}), \dots, u_N^\pi(\boldsymbol{\theta})). \quad (6)$$

In the following, we further assume that the policy parameters $\boldsymbol{\theta}$ are specified for each discrete ambient condition separately,

$$\boldsymbol{\theta} = (\theta_1, \dots, \theta_N), \quad (7)$$

and

$$\mathbf{u}^\pi(\boldsymbol{\theta}) = (u_1^\pi(\theta_1), \dots, u_N^\pi(\theta_N)). \quad (8)$$

A *separable* discrete control policy is hence parametrized by a set of parameters $\boldsymbol{\theta}$, with individual θ_n for each considered ambient condition a_n .

Remarks

- Again, θ_n can also represent a set of parameters for each ambient condition.
- In the most common setting, each θ_n simply represents the control setpoint u_n for ambient condition a_n .
- For a given control policy π , the plant model becomes a function of only the ambient condition,

$$y = f^\pi(a|\pi), \quad (9)$$

or in terms of the policy parameters,

$$y = f^\pi(a|\boldsymbol{\theta}). \quad (10)$$

- For discrete ambient conditions we write

$$f_n^\pi(\boldsymbol{\theta}) := f^\pi(a_n|\boldsymbol{\theta}), \quad (11)$$

- Similar to the plant model and ambient condition statistics we consider only stationary control policies that do not change over time.

4 Metric accumulation

We are eventually interested in *accumulated metrics* of the form

$$F = \int_0^T f(a(t), u(t)) dt, \quad (12)$$

or given a control policy π ,

$$F^\pi(\boldsymbol{\theta}) = \int_0^T f(a(t)|\boldsymbol{\theta}) dt, \quad (13)$$

where T denotes a considered time horizon, and $a(t)$ varies over time.

For given ambient condition statistics, we are interested in the expectation of $F^\pi(\boldsymbol{\theta})$,

$$\mathbb{E}[F^\pi(\boldsymbol{\theta})] = \mathbb{E}\left[\int_0^T f(a(t)|\boldsymbol{\theta})dt\right] \quad (14)$$

$$= \int_0^T \mathbb{E}[f(a(t)|\boldsymbol{\theta})] dt, \quad (15)$$

according to fundamental rules of integration and linearity of the expectation value. Due to stationarity of the ambient conditions, the expectation value is constant over time and can be taken out of the integral,

$$\begin{aligned} \mathbb{E}[F^\pi(\boldsymbol{\theta})] &= \mathbb{E}[f(a|\boldsymbol{\theta})] \int_0^T dt \\ &= T \mathbb{E}[f(a|\boldsymbol{\theta})]. \end{aligned} \quad (16)$$

The constant expectation $\mathbb{E}[f(a|\boldsymbol{\theta})]$ is computed according to the ambient condition statistics. More specifically, for the discrete ambient condition PMF we get

$$\mathbb{E}[f(a|\boldsymbol{\theta})] = \sum_{n=1}^N p_n f(a_n|\boldsymbol{\theta}), \quad (17)$$

or for separable policy parameters,

$$\mathbb{E}[f(a|\boldsymbol{\theta})] = \sum_{n=1}^N p_n f_n(\theta_n). \quad (18)$$

Finally, the expected value of the accumulated metric F^π is computed as

$$\mathbb{E}[F^\pi(\boldsymbol{\theta})] = T \sum_{n=1}^N p_n f_n(\theta_n). \quad (19)$$

Remarks

- In the following we skip the explicit mentioning of the expectation value $\mathbb{E}[\cdot]$, and will simply refer to the expected accumulated metric as F^π . Note that in practice this converges to the actual accumulated metric for large enough integration periods T .

5 Constrained Optimization problem formulation

We are interested in scenarios where two or more accumulated metrics are in conflict with each other. For simplicity, we assume a single *objective* metric F^π

which is to be maximized and results from integration of a model output f , and a single *cost* metric G^π , which must not exceed a certain threshold $G^{\max}$, and results from integration of a second model output g . The optimization problem reads

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} F^\pi(\boldsymbol{\theta}), \quad \text{s.t. } G^\pi(\boldsymbol{\theta}) \leq G^{\max}, \quad (20)$$

where the expectation values have been skipped for simplicity. Assuming a separable discrete control policy, we get

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} T \sum_{n=1}^N p_n f_n(\theta_n), \quad \text{s.t. } T \sum_{n=1}^N p_n g_n(\theta_n) \leq G^{\max}, \quad (21)$$

or

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} \sum_{n=1}^N p_n f_n(\theta_n), \quad \text{s.t. } \sum_{n=1}^N p_n g_n(\theta_n) \leq G_T^{\max}, \quad (22)$$

where the constant T has been removed from the objective expression, and

$$G_T^{\max} = \frac{G^{\max}}{T}. \quad (23)$$

Remarks

- We observe that the objective function F^π can be separated for each ambient condition n , and coupling of the full problem arises only due to the constraint on G^π .
- Note that we use the term “cost” here for an accumulated metric that is considered “bad”, which is usually not interpretable as a monetary cost. In particular, it is not a cost that can be directly subtracted from the objective function (such as e.g. operational expenditures vs. revenue), in which case the problem would readily formulated as an unconstrained optimization problem with a combined objective function.
- Again, in practice we will see a set of accumulated metric constraints rather than a single quantity, as suggested in (22). The typical example in the context of wind farm optimization would be accumulated *damage equivalent loads*, which may need to be evaluated for multiple components at every wind turbine.

6 Lagrangian Relaxation

Direct solution of (22) implies simultaneous optimization for the policy parameters $\boldsymbol{\theta}$ subject to the constraint equation(s), which would quickly result in high-dimensional and hence intractable problems. The basic idea of Lagrangian

Relaxation is to reformulate the problem as an unconstrained optimization problem that approximates the original problem, by means of Lagrangian variables λ .

We introduce the Lagrangian (assuming a single constraint for now)

$$\mathcal{L}(\boldsymbol{\theta}, \lambda) = F^\pi(\boldsymbol{\theta}) - \lambda(G^\pi(\boldsymbol{\theta}) - G_T^{\max}), \quad \lambda \geq 0, \quad (24)$$

and consider the unconstrained optimization problem of maximizing $\mathcal{L}(\boldsymbol{\theta}, \lambda)$, which yields the *dual function* of the problem,

$$q(\lambda) = \max_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \lambda) \quad (25)$$

$$= \lambda G_T^{\max} + F^\pi(\boldsymbol{\theta}) - \lambda G^\pi(\boldsymbol{\theta}). \quad (26)$$

Looking at the Lagrangian (24) we observe:

- For $\lambda > 0$, constraint violations are indeed subtracted from the objective function, similar to a virtual penalty/cost subtracted from it. As mentioned earlier, this is typically not meaningful in the considered problems, and λ can be considered a conversion factor to allow such a combination of objective and cost.
- For $\lambda = 0$, the problem reduces to the original problem, but without the constraints. Unless the problem is solved trivially (constraint is inactive at the maximum of F^π), this will result in a constraint violation.
- For increasing λ , more weight is put on satisfying the constraint than on maximizing the (original) objective function. In particular, for large λ , the constraint will typically be “over-satisfied”, in that $G^\pi(\boldsymbol{\theta}) < G_T^{\max}$ (strictly).
- Assuming a certain well-behavedness of the problem (convex $F^\pi(\boldsymbol{\theta})$, concave $G^\pi(\boldsymbol{\theta})$), there exists a unique sweet spot λ^* that results in $G^\pi(\boldsymbol{\theta}) = G_T^{\max}$ and hence exact solution of the original constrained problem (if the constraint is active).

Mathematically, we note that for any constraint-feasible $\boldsymbol{\theta}$, the Lagrangian is greater or equal to the primal objective function, $\mathcal{L}(\boldsymbol{\theta}, \lambda) \geq F^\pi(\boldsymbol{\theta})$. Further, due to

$$\max_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \lambda) \geq \mathcal{L}(\boldsymbol{\theta}^*, \lambda), \quad (27)$$

the dual function $q(\lambda)$ represents an upper bound for the primal objective function at $\boldsymbol{\theta}^*$,

$$q(\lambda) \geq F^\pi(\boldsymbol{\theta}^*), \quad (28)$$

with equality only if the constraint is exactly satisfied for $\lambda > 0$. We hence conclude that the optimal $\boldsymbol{\theta}^*$ is found

The great advantage of Lagrangian Relaxation for the given problem lies in the fact that the sum expressions in $F^\pi(\boldsymbol{\theta})$ and $G^\pi(\boldsymbol{\theta})$ can be combined in the dual function,

$$q(\lambda) = \max_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \lambda) \quad (29)$$

$$= \lambda G_T^{\max} + F^\pi(\boldsymbol{\theta}) - \lambda G^\pi(\boldsymbol{\theta}) \quad (30)$$

$$= \lambda G_T^{\max} + \sum_{n=1}^N \max_{\theta_n} p_n (f_n^\pi(\theta_n) - \lambda g_n^\pi(\theta_n)). \quad (31)$$

The task of finding The dual problem

$$\min_{\lambda \geq 0} q(\lambda) \quad (32)$$

is solved with a bisection approach using the subgradient of the dual function,

$$q'(\lambda) = G_T^{\max} - G(\boldsymbol{\theta}^*(\lambda)). \quad (33)$$

Primal problem

$$\max_{\boldsymbol{\theta}} F(\boldsymbol{\theta}) \quad (34)$$

$$\text{s.t. } G(\boldsymbol{\theta}) \leq G^{\max} \quad (35)$$

$$F(\boldsymbol{\theta}) = \sum_n p_n f(\theta_n), \quad (36)$$

$$G(\boldsymbol{\theta}) = \sum_n p_n g(\theta_n). \quad (37)$$

Lagrangian

$$\mathcal{L}(\boldsymbol{\theta}, \lambda) = F(\boldsymbol{\theta}) - \lambda(G(\boldsymbol{\theta}) - G^{\max}), \quad \lambda \geq 0. \quad (38)$$

Dual function

$$q(\lambda) = \max_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \lambda) \quad (39)$$

$$= \lambda G^{\max} + \sum_n \max_{\theta_n} p_n (f(\theta_n) - \lambda g(\theta_n)). \quad (40)$$

Dual problem

$$\min_{\lambda \geq 0} q(\lambda). \quad (41)$$

Subgradient of the dual function

$$q'(\lambda) = G^{\max} - G(\boldsymbol{\theta}^*(\lambda)). \quad (42)$$

Algorithm 1 Dual Minimization with Expansion + Bisection

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1: Choose tolerances  $\varepsilon > 0$ 
2: Choose initial stepsize  $\alpha_0 > 0$ 
3:  $\lambda \leftarrow 0$ 
▷ Expansion phase

4: repeat
5:   for all  $n$  (in parallel) do
6:      $\theta_n^*(\lambda) \leftarrow \arg \max_{\theta_n} p_n(f(\theta_n) - \lambda g(\theta_n))$ 
7:   end for
8:   Compute  $G(\theta^*(\lambda)) = \sum_n p_n g(\theta_n^*(\lambda))$ 
9:    $s(\lambda) \leftarrow G^{\max} - G(\theta^*(\lambda))$ 
10:  if  $s(\lambda) < 0$  then
11:     $\lambda \leftarrow \lambda + \alpha_0$ 
12:  end if
13: until  $s(\lambda) \geq 0$ 
14:  $\lambda_{\min} \leftarrow \lambda - \alpha_0$ 
15:  $\lambda_{\max} \leftarrow \lambda$ 
▷ Bisection phase

16: while  $\lambda_{\max} - \lambda_{\min} > \varepsilon$  do
17:    $\lambda \leftarrow \frac{1}{2}(\lambda_{\min} + \lambda_{\max})$ 
18:   for all  $n$  do
19:      $\theta_n^*(\lambda) \leftarrow \arg \max_{\theta_n} p_n(f(\theta_n) - \lambda g(\theta_n))$ 
20:   end for
21:   Compute  $G(\theta^*(\lambda))$ 
22:    $s(\lambda) \leftarrow G^{\max} - G(\theta^*(\lambda))$ 
23:   if  $s(\lambda) < 0$  then
24:      $\lambda_{\min} \leftarrow \lambda$ 
25:   else
26:      $\lambda_{\max} \leftarrow \lambda$ 
27:   end if
28: end while
29: return  $\theta^*(\lambda)$ 

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